luke@shimanuki.cc

# LUKE SHIMANUKI

https://luke.shimanuki.cc

## RECENT EXPERIENCE

**Aurora Innovation** Sr. MLE --> Staff MLE --> **Staff TLM**, Behavior ML 2022–present

- **Led the 16-engineer post-training effort** to improve driving performance of the behavior model and enable on-road deployment. Triaged and prioritized failure modes remaining after pre-training, defined the remediation roadmap, and built a repeatable SFT and DAgger workflow that enabled the team to triple in size by onboarding traditional software engineers into ML development.
- **Led 6-engineer scene understanding workstream for behavior model pre-training**, aligning planning, perception, mapping, routing, and validation teams to ensure safe on-road interactions and fault-response.
- **Led 7-engineer team to deploy motion forecasting models** enabling 500k+ incident-free driverless miles. Worked with product and downstream consumers to identify and schedule improvements, led interface and architecture design, and ensured on-time delivery unblocking 5 company-wide releases.
- **Served as primary technical point of contact for interaction behavior** in cross-org launches, demos, and incident reviews; diagnosed on-road failures, led design and delivery of fixes, and communicated timeline and scope trade-offs with senior leadership and external-facing launch/demo teams.
- **Mentored & sponsored 6 ML engineers**, creating opportunities for them to own increasing technical complexity and cross-team/leadership scope by providing design/communication guidance and acting as a safety net to encourage risk-taking; successfully advocated for 1 promotion to senior.

**Magna Electronics / Optimus Ride** SDE II --> Sr. SDE, Motion Planning 2020–2022

- Led 3 engineers across 2 teams to expand prediction-planning interface and algorithms to produce probability-aware forecasts and plans, enabling safe autonomous navigation through contested intersections.
- Formulated technical requirements & algorithmic specifications for 2 ADAS features following ISO 26262 functional safety standards, resulting in a patent filing.

## EDUCATION & RESEARCH PUBLICATIONS

**Massachusetts Institute of Technology | CSAIL Learning & Intelligent Systems** 2016–2020

**M.Eng** Electrical Engineering & Computer Science (AI Concentration), GPA 5.0 (out of 5.0)

**S.B.** Computer Science + Brain & Cognitive Science, GPA 4.9 (out of 5.0), 5.0 within CS major

B. Axelrod*, **L. Shimanuki***, "Efficient Motion Planning under Obstacle Uncertainty with Local Dependencies" WAFR 2022

**L. Shimanuki***, B. Axelrod*, "Hardness of Motion Planning with Obstacle Uncertainty in 2 Dimensions" IJRR 2021

B. Kim*, **L. Shimanuki***, et al., "Representation, learning, and planning algorithms for geometric task and motion planning" IJRR 2021

B. Kim*, **L. Shimanuki***, "Learning value functions with relational state representations for guiding task-and-motion planning" CoRL 2019

**L. Shimanuki***, B. Axelrod*, "Hardness of Motion Planning with Obstacle Uncertainty in 3 Dimensions" WAFR 2018

** indicates equal contribution*

**SKILLS** **Languages:** C, C++, Python **ML/AI:** PyTorch / TRT, TensorFlow, ROS **Tools:** Git, Docker, Linux

## EARLY CAREER

**Cruise Automation** Intern, Prediction Summer 2018

- Made forecasts dynamically feasible & intersection-aware, tripling passing tests & reducing disengagements.

**Optimus Ride (Self-driving)** SWE Intern, Perception Summer 2017

- Designed & implemented end-to-end perception-prediction pipeline from object detection through tracking & forecasting, establishing architecture that interfaces between the raw sensor data and the motion planner.

**RightHand Robotics (Grasping)** Intern, Machine Learning / Vision January 2017

- Designed & trained grasping model from RGB-D input, improving pick success rate from cluttered boxes.

**Tanius Technology (Prop Trading)** Developer, ML Modeling 2015–2016

- Built end-to-end ML system for algorithmic trading: data ingestion, model training, & backtesting infrastructure.

## ACTIVITIES

**Site Manager** *Food for Free COVID-19 Relief Program* 2020

Directed team of 12 volunteers for packing & delivering groceries to ~300 households weekly.

**Program Director, Head Webmaster** *MIT Educational Studies Program* 2017–2020

Directed educational programs reaching ~3000 MS/HS students, with ~1000 classes taught by ~500 teachers, & run by ~100 volunteers. Mentored 5 future program directors. Maintained website used by ~5000 students.

**Software Lead** *AVBotz* 2012–2016

Led the software team (~12 members) for fully-autonomous submarine for manipulating objects, aiming & shooting torpedoes, & navigating around obstacles. International finalist (7th Place) as a HS team at collegiate RoboSub 2015.

## AWARDS

**USACO Platinum** **Intel STS 2016 Semifinalist** **Eagle Scout** **MIT Battlecode 2018 Finalist**